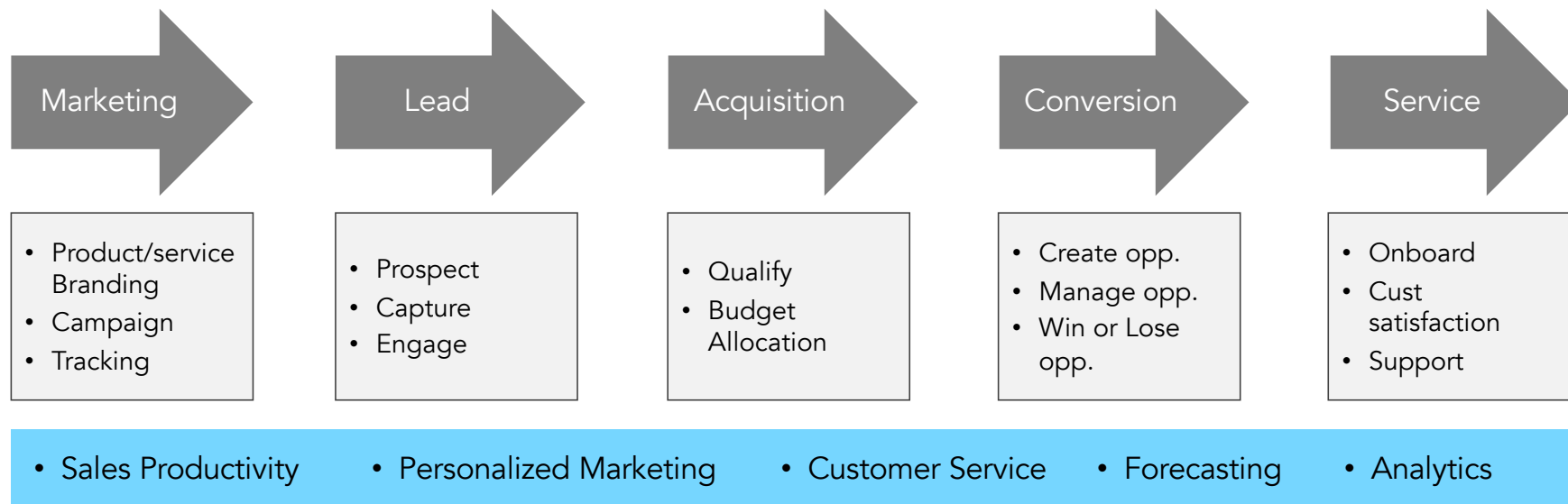


Project 12 CRM (Customer Relationship Management) Analytics

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CRM Application

- Customer Relationship Management.
- Zoho Corporation, Salesforce, Adobe CRM, Nimble, Microsoft Dynamics 365 etc.
- Used by organizations to manage customer engagement. Helps organizations to improve customer business relation, drive growth and increase revenue.

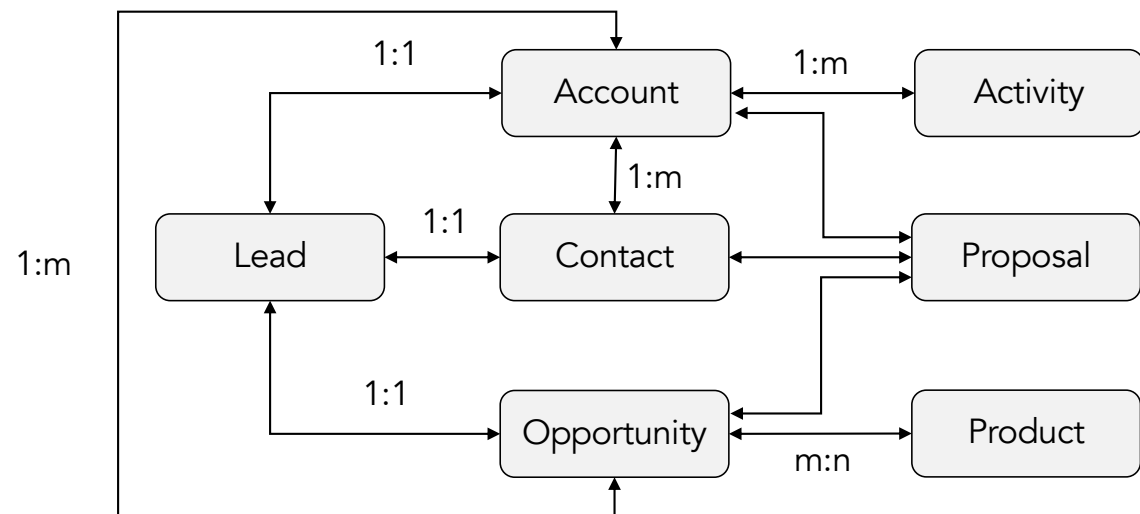


CRM Entities

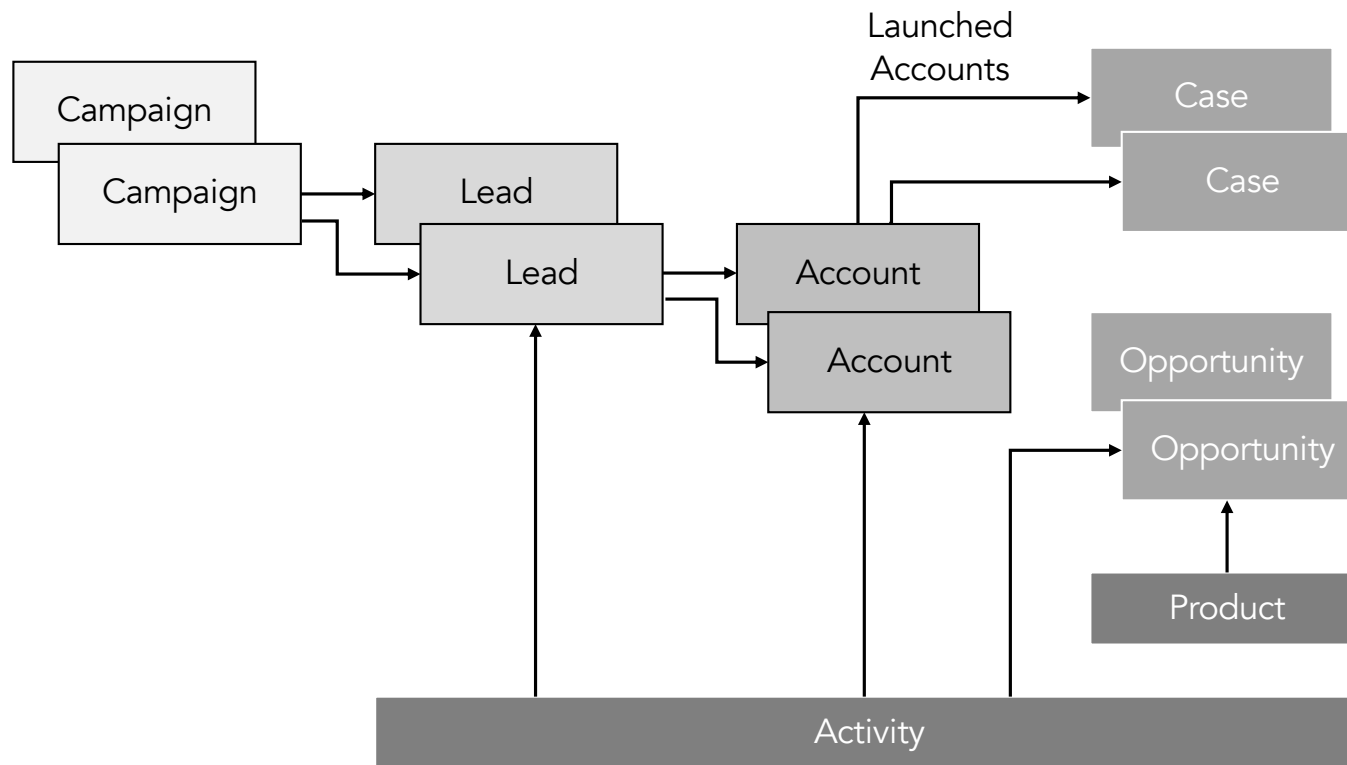


- Entities or tables

- Lead
- Account – Commit & Launch
- Product
- Contact
- Opportunity
- Activity
- Case
- Campaign

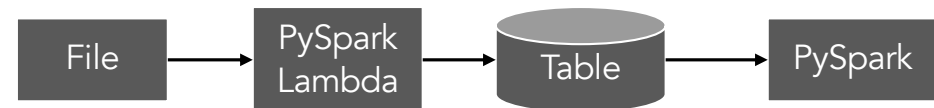


CRM Entities



Datasets

- Total Size – 38 GB (Uncompressed)
- No. of Files – 82
 - crm_activity_data_X : 1 to 20
 - crm_opportunity_data_X : 1 to 20
 - crm_account_commit_data_X : 1 to 10
 - crm_account_launch_data_X : 1 to 10
 - crm_lead_data_X : 1 to 10
 - crm_product_data : 1
 - crm_campaign_data : 1
 - crm_case_data_X : 1 to 10



Entity Details

crm_lead

Column	Description
lead_id	System generated
customer_org_name	Org name for CRM product
customer_org_sector	Private, Public, FinTech, Finance, Auto, NPO, Govt, Enterprise, DigitalNative, Startup, LLC, InvestmentBank, RetailBank, ConsumerRetail, FoodTech, Games, Education, PublicSectorFinance, PublicDefence, PrivateDefence, GovtThinkTank, PrivateThinkTank, PrivateHospital, PublicHealth.
lead_source	From where lead was generated
lead_status	Current status of the lead. Unqualified → qualified → verification → bus-validation → tech-validation → committed → launched
sales_rep	Unique id of sales representative
sales_rep_country	Country of sales rep
created_date	When the lead was created
conversion_date	Status change of the lead

crm_product

Column	Description
product_id	Unique ID
product_category	'Product', 'Service', 'OnDemandTechnology'
product_name	'FinanceRoll', 'PaymentGateway', 'ClaimProcessing', 'MobiPay', 'eWallet', 'RegTech'. 'Development', 'Testing', 'DBA', 'DataEngineer', 'MLEngineer'. 'PaaS', 'IaaS', 'SaaS', 'Serverless'
product_unit_sku	LicensePerUser, ManHours, PricePerHour
product_unit_price	Based on category and name.

Entity Details

crm_opportunity

Column	Description
opp_id	Unique identifier
opp_name	Name of the opp
acc_id	Account the opp is linked with
acc_open_date	When opportunity was opened in CRM
acc_close_date	opportunity close date
opp_product_id	Product used in the opp
opp_product_unit	LicensePerUser, ManHours, PricePerHour
opp_product_spent	No. of licenses No. of man-hours No. of hours spent in the opp
opp_status	Commit or Launched

crm_account

Column	Description
account_id	Unique ID
account_name	Org name for CRM product
account_industry	Private, Public, FinTech, Finance, Auto, NPO, Govt, Enterprise, DigitalNative, Startup, LLC, InvestmentBank, RetailBank, ConsumerRetail, FoodTech, Games, Education, PublicSectorFinance, PublicDefence, PrivateDefence, GovtThinkTank, PrivateThinkTank, PrivateHospital, PublicHealth.
account_poc_start_date (only for Commit account)	Start date of POC.
account_owner	Based on category and name. Sales Rep of Opp table.
account_launch_date (only for Launched account)	When the account was launched.
account_projected_annual_revenue (only for Launched account)	Projected revenue of the account.

Entity Details

crm_campaign

Column	Description
campaign_id	Unique ID
campaign_type	MarketingEvent, DirectOutreach, TradeShow, WebForm, EmailMarketing, OnlineAd, Referral, OnlineContent, UserEvent, DeveloperEvent, PaperAd, AppMarketplace
campaign_date	Date of event.
campaign_cost	Money spent on the campaign event.

crm_activity

Column	Description
activity_id	Unique ID with Link Opp ID.
link_opp_id	Opp ID for this activity.
lead_acc_opp_flag	"L" or "A" or "O". If "O" – Use link_opp_id to link activity with Opportunity. If "A" – Use acc_id with opp_id to link activity with Account. If "L" – Use acc_id from opp table and acc_owner from account table to link activity with Lead.
activity_type	"email", "meeting", "phone", "code", "unit-test", "integration-test", "design", "planning", "architecture", "troubleshoot", "break-fix"
activity_status	"open", "closed", "cancelled", "acknowledged", "working"
activity_open_date	Activity opened.
activity_close_date	Activity closed.

Entity Details

crm_case

Column	Description
case_id	Unique ID.
case_status	New, Working, L1, L2, L3, L4 Escalated, Waiting on Customer, Waiting on Support, Closed.
case_priority	P0, P1, P2, P3, P4.
case_account_id	Account to which the case is related to.
case_origin	Ticket, Email, Chat, Phone, Text.
case_sla	1 to 15 days.
case_rating	Rating provided by customer on CLOSED cases. 1 (Lowest) to 10 (Highest).

Problem Statements 1

1. Which are the top 5 products that had the highest no of POCs and highest spent during the POCs.
2. Top 5 products that are used the most across different sectors, month-on-month (MOM).
3. What is the total product spent (across all products) attainment MOM against projected revenue.
$$\text{Attainment} = (\text{product spent YTD}) * 100 / ((\text{projected revenue of acc}/12) * \text{no_of_months considered in YTD}).$$
4. Total spent for each product over the years (Year-on-Year) across customer sectors.
5. Distribution of product category across account industries.
6. Average POC days (opp_product_id of opportunity table when opp_status is "commit" - account_poc_start_date of account table) for each product.
7. Which customers are exploring Cloud technologies (Category - OnDemandTechnology) in their POCs.
8. What is the month-on-month attainment against projected revenue of each account over the years.
9. Which top 3 product and their categories are most famous each year based on revenue. Also, country-wise.
10. Which country uses the highest "Service" based product each year?

Problem Statements 2

11. What budget was spent on campaign each year and what is its no. (count) of lead generation for those years.
12. What is the Return on Investment of Campaigns each year.
 - $\text{Return on Investment (ROI)} = \text{Revenue from "Launched" opportunities each year} / \text{Total campaign cost that year.}$
 - $\text{Revenue from "Launched" opp.} = \text{Total amount earned from all categories of products each year.}$
13. What is the avg. duration of leads at each stage (Unqualified, Qualified, Verification etc.) ?
14. What are the top 3 lead sources each year and their corresponding revenue realization (Same as revenue from "launched" opp.) ?
15. What is the avg. no. of POCs and their duration for each account.
16. What is top 2 most famous campaign each year and their conversion rate (No. of leads for lead source similar to campaign type).
17. What is the highest activity for each account, opportunity and lead and their corresponding duration.
18. Which activity is performed the most during POC stage of an account.
19. What is the avg. time spent from planning to testing (plan, design, architect, code, unit-test)

Problem Statements 3

20. Which products (product_category is Product or OnDemandTechnology) are used the most in design and architecture (activity type = design and architecture) phases.
21. Which account industry uses cloud services ('PaaS','IaaS','SaaS','Serverless') the most.
22. For each account, what is the total yearly revenue generation as opposed to projected revenue over the years and their corresponding account owners.
23. Which product unit SKU type generate highest revenue each year for all launched accounts.
24. What is the longest running account for each sector and their account owners.
25. What is the highest, avg. and lowest rating provided by customers per industry.
26. Which products (product_name) have the lowest rating provided by customers.
27. Which individual products (product_name) have the highest support cases each year.
28. What is the percentage of support cases that have been gone to L4 engineer or Escalated for individual products (product_name).
29. What is the distribution of origin of support case across each products.
30. Which products have the highest no. of opportunities across sectors.
31. Which "service" is used the most across various customer sectors. It is needed for head count or resource management.